Dataset with brain atlas A: use the Oasis dataset **for classification** of male and female individuals based on structural brain scans

```
import numpy as np
from nilearn import datasets
from nilearn.input_data import NiftiLabelsMasker
from nilearn.image import index_img
import nibabel as nib

brain_data = datasets.fetch_oasis_vbm(n_subjects=100)
yeo = datasets.fetch_atlas_schaefer_2018(n_rois=100)  # this needs to be modified for some tasks
print('%i regions found in this atlas' % (len(np.unique(nib.load(yeo.maps).get_data().ravel())) - 1))

masker = NiftiLabelsMasker(labels_img=yeo.maps, standardize=True,
    memory='nilearn_cache')
input_variables = masker.fit_transform(brain_data.gray_matter_maps)
output_variable = np.array(brain_data.ext_vars.mf == 'F', dtype=int)  # gives 1 for females and 0 for males
```

Task 1- Build a learning curve. Using LogisticRegression(penalty='l2', C=1) and the lowest resolution data (nrois=100), train a classification model on different subsets of the 100 subjects in increments of 10 (ie. Always using the 100 brain region features, and 10,20,…,90 subjects – you will have 9 different models). Next, evaluate each fitted LogReg model on the unseen last 10 subject brain scans (these should be the same 10 subjects across models). Plot the classification accuracy on the training set and the test set (x axis=number of subjects (10-90), y axis=classification accuracy, 2 colors for two lines for the 2 different metrics).

Task 2- Same as 1 but using the highest atlas resolution (nrois=1000).

Task 3- Building on task 1+2, plot other performance metrics instead of classification accuracy. Using the same train/test set as Q1/2 with 100 ROIs (nrois=100), plot the following test performance metrics: a) precision, b) recall, c) f1 score, d) sensitivity, and e) specificity (5 lines with 5 colors, same axes as Q1/2).

Task 4- Confusion matrix: similar to task 1 but using k-fold (k=10) cross-validation on 100 brain regions (nrois=100) to produce the characteristic 2x2 confusion matrix of actual-vs-predicted label (model label prediction on test set vs true test set label). Aggregate the test set predictions across the ten folds so that you have 100 (predicted label, true label) pairs.

Task 5- Calibration plot: Similar to task 1 but using k-fold (k=10) cross-validation on 100 brain regions to produce a calibration plot (requires predict_proba() function of LogReg(), implemented in sklearn, looks like in lecture).

Dataset with brain atlas B: use the Oasis dataset to predict the **continuous** age based on structural brain scans

```
import numpy as np
from nilearn import datasets
from nilearn.input_data import NiftiLabelsMasker
from nilearn.image import index_img
from sklearn.preprocessing import StandardScaler
import nibabel as nib

brain_data = datasets.fetch_oasis_vbm(n_subjects=100)
crad = datasets.fetch_atlas_craddock_2012()
atlas_nii = index_img(crad['scorr_mean'], 42) # this needs to be modified for some tasks
print('%i regions found in this atlas' % (len(np.unique(atlas_nii.get_data().ravel())) - 1))

masker = NiftiLabelsMasker(labels_img= atlas_nii, standardize=True)
input_variables = masker.fit_transform(brain_data.gray_matter_maps)
output_variable = StandardScaler().fit_transform(brain_data.ext_vars.age[:, None])[:, 0] #
gives subject age on standard units after z-scoring
```

Note: use Ridge() instead of LogisticRegression(), and coefficient of determination (sklearn.metrics.r2_score) instead of accuracy.

6- analogous to task 1 but predicting continuous subject age and using the 1st / lowest-resolution version of Craddock region atlas resolutions (.scorr_mean).

7- analogous to task 2 but predicting continuous subject age and using the last / finest-grained-resolution version of Craddock region atlas resolutions (.scorr_mean).

8- analogous to task 3 but plotting outcome metrics suited to continuous outcomes: a) r2 score, b) mean absolute error and c) explained variance score, as well as d) mean-squared error.

9- analogous to task 4, median-splitting the continuous outcome (that is, computing median of outcome across ages on the full sample of participants, then classifying predictions into high-age vs. low-age on the test participants), then produce a confusion matrix in similar manner for a 2x2 confusion matrix of actual vs. predicted high- vs. low-age.

10- analogous to task 5, but using age prediction (since the target is continuous, divide into high-age vs. low-age groups and use LogisticRegression() as in Q5).